

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2022 P2H Worked Solutions

Jun 2022 | P2H

29 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

- 1** Here are some integers where $a < b < c < d$

$a \quad b \quad c \quad d \quad d \quad d$

The mode of the integers is 9

The median of the integers is 8

The range of the integers is 4

Work out the value of a , the value of b , the value of c and the value of d

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots$

$d = \dots\dots\dots$

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Calculate statistic

The mode is 9, so

$$d = 9$$

2. The range is

The range is 4:

$$9 - a = 4$$

$$a = 5$$

3. Calculate statistic

The median is 8, so

$$\frac{c + d}{2} = 8$$

$$\frac{c + 9}{2} = 8$$

$$c = 7$$

4. The integers are in orderThe integers are in order and $a < b < c$, so

$$b = 6$$

FINAL ANSWER $a = 5, b = 6, c = 7, d = 9.$

PAST PAPER QUESTION**Q#: 2****Marks: 4****TOPIC: Graphing Inequalities**

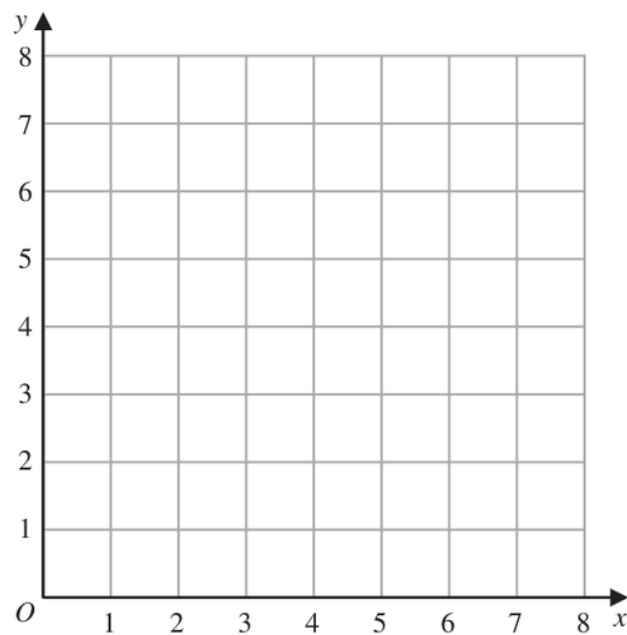
Jun 2022 P2H - Jun 2022 | P2H

2 (a) On the grid, draw and label with its equation the straight line with equation

(i) $y = 1$

(ii) $x = 2$

(iii) $x + y = 7$



(3)

(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$y \geq 1$

$x \geq 2$

$x + y \leq 7$

Label the region **R**.

(1)

(Total for Question 2 is 4 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 4****TOPIC: Graphing Inequalities**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Draw the boundary lines**

Draw the boundary lines:

$$y = 1, \quad x = 2, \quad x + y = 7$$

2. The required region is

The required region is

$$y \geq 1$$

$$x \geq 2$$

$$x + y \leq 7$$

FINAL ANSWERShade $y \geq 1$, $x \geq 2$, $x + y \leq 7$.

PAST PAPER QUESTION**Q#: 3****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

- 3** An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

(Total for Question 3 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 3 Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. The time is

The time is

$$5 \text{ h } 24 \text{ min} = 5.4 \text{ h}$$

2. Average speed

Average speed:

$$3980 \div 5.4 = 737.037 \dots$$

FINAL ANSWER

737 kilometres per hour.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 3****TOPIC: Fractions**

Jun 2022 P2H - Jun 2022 | P2H

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

(Total for Question 4 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 4****Marks: 3****TOPIC: Fractions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify fraction**

$$5\frac{1}{3} = \frac{16}{3}$$

$$2\frac{6}{7} = \frac{20}{7}$$

2. Use a common denominator

Use a common denominator:

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21}$$

$$= \frac{52}{21} = 2\frac{10}{21}$$

FINAL ANSWER

$$2\frac{10}{21}$$

PAST PAPER QUESTION**Q#: 5****Marks: 4****TOPIC: Area & Perimeter**

Jun 2022 P2H - Jun 2022 | P2H

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

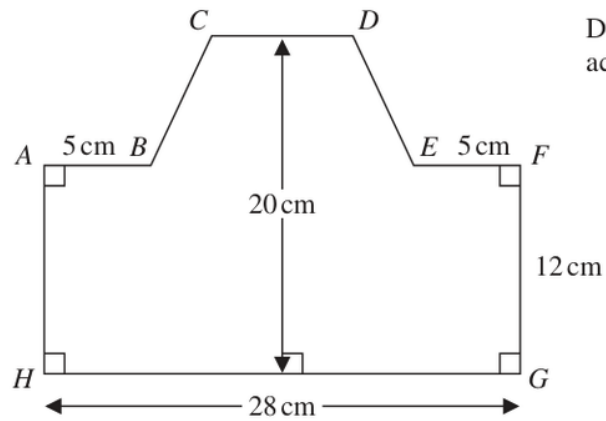


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 4****TOPIC: Area & Perimeter**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The lower rectangle has width**

The lower rectangle has width 28 cm and height 12 cm:

$$28 \times 12 = 336 \text{ cm}^2$$

2. The top part is a

The top part is a trapezium of height

$$20 - 12 = 8 \text{ cm}$$

3. Find the gradient

The lower parallel side of this trapezium is

$$28 - 5 - 5 = 18 \text{ cm}$$

4. Area of the trapezium

Area of the trapezium:

$$434 - 336 = 98 \text{ cm}^2$$

5. Use

Use

$$\text{area} = \frac{1}{2}(a + b)h$$

$$98 = \frac{1}{2}(18 + CD)(8)$$

$$98 = 4(18 + CD)$$

$$18 + CD = 24.5$$

$$CD = 6.5$$

FINAL ANSWER

6.5 cm.

PAST PAPER QUESTION**Q#: 6****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2022 P2H - Jun 2022 | P2H

- 6 The diagram shows triangle PQR .

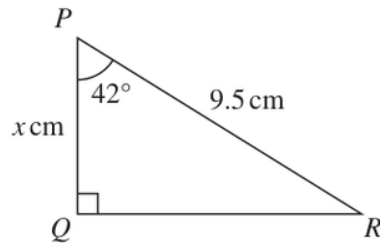


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

(Total for Question 6 is 3 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**

In the right-angled triangle, x is adjacent to the 42° angle and 9.5 cm is the hypotenuse.

$$\cos 42^\circ = \frac{x}{9.5}$$

$$x = 9.5 \cos 42^\circ = 7.059 \dots$$

FINAL ANSWER

7.1 cm.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

(Total for Question 7 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Standard & Compound Units**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION

Dr. Eslam Ahmed

1. Convert km/h to m/s divide

To convert km/h to m/s, divide by 3.6.

$$81 \div 3.6 = 22.5$$

FINAL ANSWER

22.5 m/s.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 5****TOPIC: Percentages**

Jun 2022 P2H - Jun 2022 | P2H

8 Behnaz makes 300 celebration cards so that

number of birthday cards : number of anniversary cards : number of congratulations cards = 7:5:3

$\frac{2}{5}$ of the birthday cards have numbers on them.

36% of the anniversary cards have numbers on them.

None of the congratulations cards have numbers on them.

Work out what fraction of the 300 cards have numbers on them.

Give your answer in its simplest form.

(Total for Question 8 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 8****Marks: 5****TOPIC: Percentages**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The ratio has parts**

The ratio 7 : 5 : 3 has 15 parts.

2. Birthday cards

Birthday cards:

$$\frac{7}{15} \times 300 = 140$$

3. Anniversary cards

Anniversary cards:

$$\frac{5}{15} \times 300 = 100$$

4. Congratulations cards

Congratulations cards:

$$\frac{3}{15} \times 300 = 60$$

5. Birthday cards with numbers

Birthday cards with numbers:

$$\frac{2}{5} \times 140 = 56$$

6. Anniversary cards with numbers

Anniversary cards with numbers:

$$36\% \times 100 = 36$$

7. Congratulations cards with numbers

Congratulations cards with numbers:

$$0$$

8. Total with numbers

Total with numbers:

$$56 + 36 = 92$$

9. Fraction of all cards

Fraction of all cards:

$$\frac{92}{300} = \frac{23}{75}$$

FINAL ANSWER

$$\frac{23}{75}$$

EA

PAST PAPER QUESTION

Q#: 9

Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2022 P2H - Jun 2022 | P2H

- 9 Pasha invests 50 000 dollars in a savings account for 4 years.
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.
Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 9 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 9****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The multiplier for compound interest**

The multiplier for 1.3% compound interest is

$$1.013$$

2. Use compound interest

After 4 years,

$$50000(1.013)^4 = 52651.1408\dots$$

3. Use compound interest

Correct to the nearest dollar,

$$52651.1408\dots \approx 52651$$

FINAL ANSWER

52651 dollars

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2022 P2H - Jun 2022 | P2H

10 Solve the simultaneous equations

$$\begin{aligned}7x + 3y &= 3 \\ 3x - y &= 7\end{aligned}$$

Show clear algebraic working.

 $x = \dots\dots\dots$ $y = \dots\dots\dots$

(Total for Question 10 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 10****Marks: 3****TOPIC: Simultaneous Equations**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Solve simultaneous equations**

$$7x + 3y = 3$$

$$3x - y = 7$$

2. From the second equation

From the second equation:

$$y = 3x - 7$$

3. Substitute into the first equation

Substitute into the first equation:

$$7x + 3(3x - 7) = 3$$

$$7x + 9x - 21 = 3$$

$$16x = 24$$

$$x = \frac{3}{2}$$

4. Solve simultaneous equations

Then

$$y = 3\left(\frac{3}{2}\right) - 7 = \frac{9}{2} - 7 = -\frac{5}{2}$$

FINAL ANSWER

$$x = \frac{3}{2}, y = -\frac{5}{2}.$$

PAST PAPER QUESTION**Q#: 11****Marks: 3****TOPIC: Factorising**

Jun 2022 P2H - Jun 2022 | P2H

11 (i) Factorise $x^2 + 5x - 24$
(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$
(1)

(Total for Question 11 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Factorising**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Factorise expression**

$$x^2 + 5x - 24 = (x + 8)(x - 3)$$

2. Factorise expression

So

$$(x + 8)(x - 3) = 0$$

FINAL ANSWER

$$x = -8 \text{ or } x = 3$$

EA

PAST PAPER QUESTION

Q#: 12

Marks: 3

TOPIC: **Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

12 Larry is a delivery man.

He has 7 parcels to deliver.

The mean weight of the 7 parcels is 2.7 kg

Larry delivers 3 of the parcels.

Each of these 3 parcels has a weight of W kg

The mean weight of the other 4 parcels is 3.3 kg

Work out the value of W

$W =$

(Total for Question 12 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 12****Marks: 3****TOPIC: Statistics Toolkit**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Total mass of the 7**

Total mass of the 7 parcels:

$$7 \times 2.7 = 18.9$$

2. Total mass of the 4

Total mass of the 4 remaining parcels:

$$4 \times 3.3 = 13.2$$

3. Total mass of the 3

Total mass of the 3 delivered parcels:

$$18.9 - 13.2 = 5.7$$

4. Each delivered parcel has massEach delivered parcel has mass W :

$$3W = 5.7$$

$$W = 1.9$$

FINAL ANSWER

1.9 kg.

PAST PAPER QUESTION

Q#: 13 Marks: 7

TOPIC: Cumulative Frequency Diagrams

Jun 2022 P2H - Jun 2022 | P2H

13 The table gives information about the ages, in years, of 80 people in a train carriage.

Age (a years)	Frequency
$0 < a \leq 20$	7
$20 < a \leq 30$	25
$30 < a \leq 40$	20
$40 < a \leq 50$	14
$50 < a \leq 60$	8
$60 < a \leq 70$	6

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

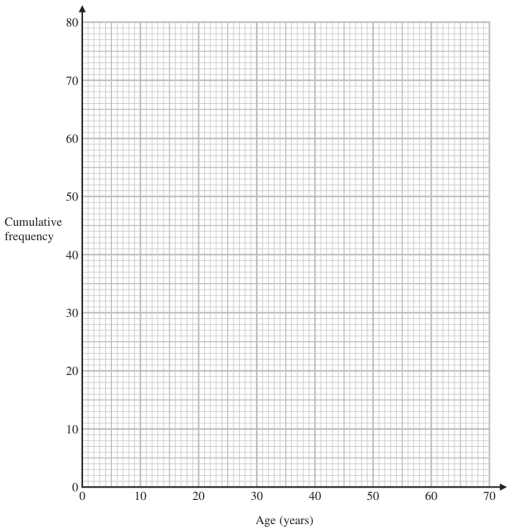
(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

_____ years
(1)



Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

(d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

(3)

(Total for Question 13 is 7 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 7****TOPIC: Cumulative Frequency Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Complete the cumulative frequencies**

Complete the cumulative frequencies:

7, 32, 52, 66, 74, 80

2. Plot the points

Plot the points

(0, 0), (20, 7), (30, 32), (40, 52), (50, 66), (60, 74), (70, 80)

3. The median is at cumulative

The median is at cumulative frequency 40.

4. Convert standard form

This is between (30, 32) and (40, 52):

$$30 + \frac{40 - 32}{52 - 32} \times 10 = 34$$

5. Use cumulative frequency

So the median age is about 34 years.

6. Estimate the number aged between

Estimate the number aged between 18 and 65:

$$F(18) \approx \frac{18}{20} \times 7 = 6.3$$

$$F(65) \approx 74 + \frac{5}{10}(80 - 74) = 77$$

$$77 - 6.3 = 70.7$$

7. 60% are going to work

60% are going to work:

$$0.6(70.7) = 42.42$$

FINAL ANSWER

cumulative frequencies 7, 32, 52, 66, 74, 80, median about 34 years, about 42 people going to work.

PAST PAPER QUESTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2022 P2H - Jun 2022 | P2H

- 14** (a) Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

.....
(3)

- (b) Make c the subject of $g = \frac{c + 3}{4 + c} - 7$

.....
(4)

(Total for Question 14 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$(5 - x)(2x + 3)(x + 4)$$

$$(2x + 3)(x + 4) = 2x^2 + 11x + 12$$

$$(5 - x)(2x^2 + 11x + 12) = -2x^3 - x^2 + 43x + 60$$

2. (b)

(b)

$$g = \frac{c + 3}{4 + c} - 7$$

$$g + 7 = \frac{c + 3}{c + 4}$$

$$(g + 7)(c + 4) = c + 3$$

$$c(g + 6) = -4g - 25$$

FINAL ANSWER

$$(a) -2x^3 - x^2 + 43x + 60, (b) c = \frac{-4g - 25}{g + 6}$$

PAST PAPER QUESTION**Q#: 15****Marks: 7****TOPIC: Solving Linear Equations**

Jun 2022 P2H - Jun 2022 | P2H

15 (a) Solve $\frac{4x+5}{3} - \frac{3-2x}{2} = 13$

Show clear algebraic working.

$$x = \dots\dots\dots$$

(4)

(b) Solve the inequality $2y^2 - 7y - 30 \leq 0$
Show your working clearly.

$$\dots\dots\dots$$

(3)

(Total for Question 15 is 7 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 7****TOPIC: Solving Linear Equations**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$$

2. Multiply by 6

Multiply by 6:

$$2(4x + 5) - 3(3 - 2x) = 78$$

$$8x + 10 - 9 + 6x = 78$$

$$14x + 1 = 78$$

$$14x = 77$$

$$x = \frac{77}{14} = \frac{11}{2}$$

3. (b)

(b)

$$2y^2 - 7y - 30 \leq 0$$

4. Factorise

Factorise:

$$(2y + 5)(y - 6) \leq 0$$

5. Solve quadratic equationThe roots are $-\frac{5}{2}$ and 6. The quadratic is ≤ 0 between the roots.**FINAL ANSWER**

$$x = \frac{11}{2}, \text{ and } -\frac{5}{2} \leq y \leq 6.$$

PAST PAPER QUESTION**Q#: 16** **Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

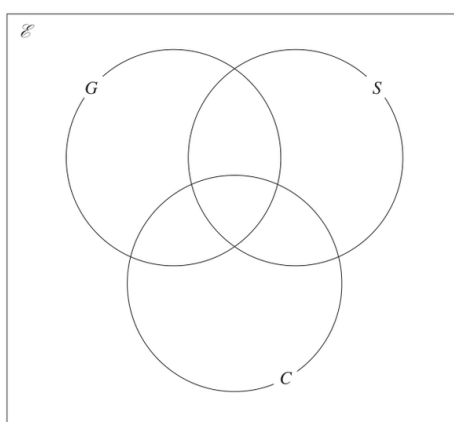
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

- (a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.



(3)

- (b) Find

(i) $n(G)$

(1)

(ii) $n([G \cup S]')$

(1)

(iii) $n(G' \cap C)$

(1)

One of the farmers who has chickens is chosen at random.

- (c) Find the probability that this farmer also has goats.

(2)

(Total for Question 16 is 8 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 8****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are farmers and have**

There are 100 farmers and 20 have none of the three animals, so 80 are in the Venn diagram.

2. Pair only regions

Pair-only regions:

$$S \cap G \text{ only} = 11 - 6 = 5$$

$$S \cap C \text{ only} = 17 - 6 = 11$$

$$G \cap C \text{ only} = 18 - 6 = 12$$

$$S \text{ only} = 31 - 5 - 11 - 6 = 9$$

$$C \text{ only} = 53 - 11 - 12 - 6 = 24$$

3. Calculate probability

Then

$$G \text{ only} = 80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$$

$$n(G) = 13 + 5 + 12 + 6 = 36$$

$$n((G \cup S)') = 20 + 24 = 44$$

$$n(G' \cap C) = 24 + 11 = 35$$

$$P(G | C) = \frac{18}{53}$$

FINAL ANSWER

36, 44, 35, $\frac{18}{53}$.

PAST PAPER QUESTION

Q#: 17

Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jun 2022 P2H - Jun 2022 | P2H

17 M varies directly as the cube of h
 $M = 4$ when $h = 0.5$

Find the value of h when $M = 500$

(Total for Question 17 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 17****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function** M varies directly as h^3 , so

$$M = kh^3$$

2. Use whenUse $M = 4$ when $h = 0.5$:

$$4 = k(0.5)^3$$

$$k = \frac{4}{0.125} = 32$$

3. Find inverse function

So

$$M = 32h^3$$

4. Find inverse functionWhen $M = 500$,

$$500 = 32h^3$$

$$h^3 = 15.625$$

$$h = 2.5$$

FINAL ANSWER

$$h = 2.5.$$

PAST PAPER QUESTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2H - Jun 2022 | P2H

18 $X = \frac{2a - b}{f}$

 $a = 7.5$ correct to 1 decimal place. $b = 3.42$ correct to 2 decimal places. $f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X
Show your working clearly.

(Total for Question 18 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 3****TOPIC: Rounding, Estimation & Bounds**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Estimate the value**

$$X = \frac{2a - b}{f}$$

2. Find upper bound

For the upper bound, make the numerator large and the denominator small.

$$a < 7.55, \quad b \geq 3.415, \quad f \geq 1.5$$

$$X_{\text{upper}} = \frac{2(7.55) - 3.415}{1.5} = 7.79$$

FINAL ANSWER

7.79

PAST PAPER QUESTION**Q#: 19****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2022 P2H - Jun 2022 | P2H

19 $a = \frac{14}{3x-7}$ $x = \frac{7}{4y-3}$

Express a in the form $\frac{py+q}{ry+s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a =$

(Total for Question 19 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 19****Marks: 3**TOPIC: **Algebraic Fractions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify fraction**

$$a = \frac{14}{3x-7}, \quad x = \frac{7}{4y-3}$$

2. Simplify fractionSubstitute for x :

$$\begin{aligned} 3x - 7 &= \frac{21}{4y-3} - 7 \\ &= \frac{21 - 7(4y-3)}{4y-3} = \frac{42 - 28y}{4y-3} \end{aligned}$$

3. Simplify fraction

So

$$a = \frac{14}{\frac{42-28y}{4y-3}} = \frac{4y-3}{3-2y}$$

FINAL ANSWER

$$a = \frac{4y-3}{3-2y}$$

PAST PAPER QUESTION

Q#: 20 Marks: 4

TOPIC: **Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

20 The diagram shows four identical circles drawn inside a square.

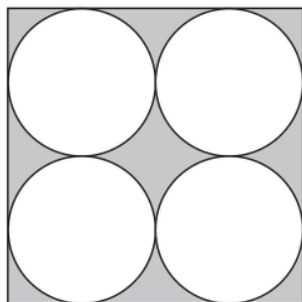


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the radius of each**Let the radius of each circle be r .**2. There are two circles across**

There are two circles across the width of the square, so the side length of the square is

$$4r$$

3. Area of the square

Area of the square:

$$(4r)^2 = 16r^2$$

4. Area of the four circles

Area of the four circles:

$$4\pi r^2$$

5. Calculate areaThe shaded area is 40 cm^2 , so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131 \dots$$

6. The perimeter of the square

The perimeter of the square is

$$4(4r) = 16r$$

$$16r = 54.6101 \dots$$

FINAL ANSWER

54.6 cm

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

20 The diagram shows four identical circles drawn inside a square.

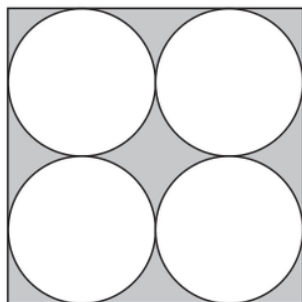


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Circles, Arcs & Sectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Let the radius of each**Let the radius of each circle be r .**2. There are two circles across**

There are two circles across the width of the square, so the side length of the square is

$$4r$$

3. Area of the square

Area of the square:

$$(4r)^2 = 16r^2$$

4. Area of the four circles

Area of the four circles:

$$4\pi r^2$$

5. Calculate areaThe shaded area is 40 cm^2 , so

$$16r^2 - 4\pi r^2 = 40$$

$$r^2(16 - 4\pi) = 40$$

$$r = \sqrt{\frac{40}{16 - 4\pi}} = 3.4131 \dots$$

6. The perimeter of the square

The perimeter of the square is

$$4(4r) = 16r$$

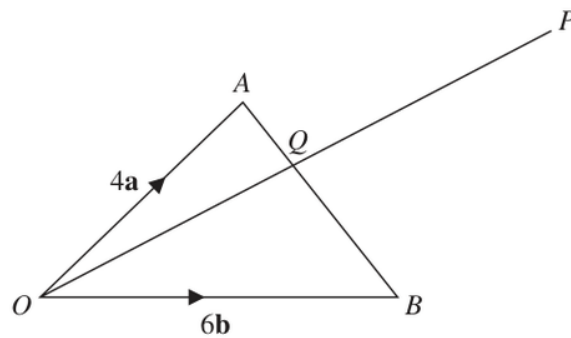
$$16r = 54.6101 \dots$$

FINAL ANSWER

54.6 cm

PAST PAPER QUESTION**Q#: 21****Marks: 5****TOPIC: Vectors**

Jun 2022 P2H - Jun 2022 | P2H

21Diagram **NOT**
accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 6\mathbf{b} \quad \vec{AP} = 2\mathbf{a} + 8\mathbf{b}$$

Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Split the ratio**

$$\overrightarrow{OA} = 4a, \quad \overrightarrow{OB} = 6b, \quad \overrightarrow{AP} = 2a + 8b$$

2. Split the ratio

So

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 4a + (2a + 8b) = 6a + 8b$$

3. Split the ratioSince Q lies on OP , let

$$\overrightarrow{OQ} = \lambda(6a + 8b)$$

4. Split the ratioSince Q lies on AB , let

$$\overrightarrow{OQ} = \overrightarrow{OA} + \mu\overrightarrow{AB}$$

$$\overrightarrow{AB} = 6b - 4a$$

5. Split the ratio

So

$$\overrightarrow{OQ} = 4a + \mu(6b - 4a) = (4 - 4\mu)a + 6\mu b$$

6. Equating coefficients

Equating coefficients:

$$6\lambda = 4 - 4\mu, \quad 8\lambda = 6\mu$$

7. Split the ratioFrom $8\lambda = 6\mu$, $\mu = \frac{4}{3}\lambda$.

$$6\lambda = 4 - \frac{16}{3}\lambda$$

$$34\lambda = 12, \quad \lambda = \frac{6}{17}$$

$$\mu = \frac{8}{17}$$

8. Split the ratio

So $AQ : QB = \frac{8}{17} : \frac{9}{17}$.

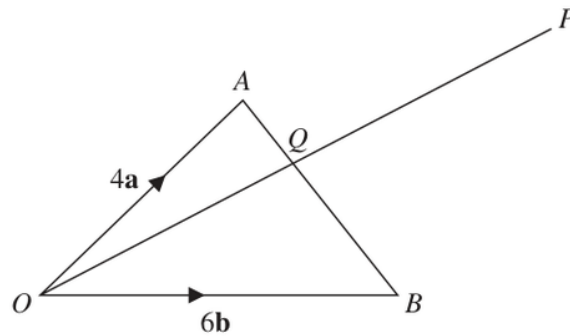
FINAL ANSWER

$$AQ : QB = 8 : 9$$

EA

PAST PAPER QUESTION**Q#: 21****Marks: 5**TOPIC: **Vectors**

Jun 2022 P2H - Jun 2022 | P2H

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accurately drawn OAB is a triangle. Q is the point on AB such that OQP is a straight line.

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Using a vector method, find the ratio $AQ:QB$

$$AQ:QB = \dots\dots\dots$$

(Total for Question 21 is 5 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 5****TOPIC: Vectors**

Jun 2022 P2H - Jun 2022 | P2H

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So

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FINAL ANSWER

$$AQ : QB = 8 : 9$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 7****TOPIC: Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

22 $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(.....,)

(.....,)

(Total for Question 22 is 7 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 7****TOPIC: Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The midpoint of is**The midpoint of BD is $M = (0, 2)$.**2. Find the gradient**Since $A = (-3, 4)$, the gradient of AM is

$$\frac{2 - 4}{0 - (-3)} = -\frac{2}{3}$$

3. Find the gradientFor the kite, diagonal AC is perpendicular to BD , so the gradient of BD is

$$\frac{3}{2}$$

4. Use direction vector alongUse direction vector $(2, 3)$ along BD .**5. Let**

Let

$$B = (2t, 2 + 3t)$$

6. Calculate value

and

$$D = (-2t, 2 - 3t)$$

7. Calculate valueSince $AB = 6.5$,

$$(2t - (-3))^2 + (2 + 3t - 4)^2 = 6.5^2$$

$$(2t + 3)^2 + (3t - 2)^2 = 42.25$$

$$4t^2 + 12t + 9 + 9t^2 - 12t + 4 = 42.25$$

$$13t^2 + 13 = 42.25$$

$$13t^2 = 29.25$$

$$t^2 = 2.25$$

$$t = 1.5$$

8. Use

The x -coordinate of B is positive, so use $t = 1.5$.

$$B = (3, 6.5)$$

$$D = (-3, -2.5)$$

FINAL ANSWER

$$B = (3, 6.5), D = (-3, -2.5)$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 7****TOPIC: Coordinate Geometry**

Jun 2022 P2H - Jun 2022 | P2H

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(..... ,)

(Total for Question 22 is 7 marks)

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Jun 2022 P2H - Jun 2022 | P2H

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FINAL ANSWER

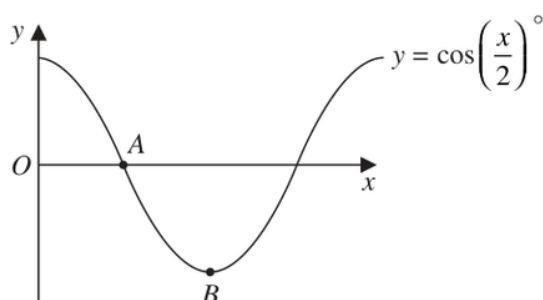
$$B = (3, 6.5), D = (-3, -2.5)$$

EA

PAST PAPER QUESTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. The graph is**

The graph is

$$y = \cos\left(\frac{x}{2}\right)^\circ$$

2. Point is the first positivePoint A is the first positive x -intercept, so $y = 0$.

$$\cos\left(\frac{x}{2}\right) = 0$$

$$\frac{x}{2} = 90^\circ$$

$$x = 180$$

3. Read the graph

So

$$A = (180, 0)$$

4. Point is the minimum pointPoint B is the minimum point.**5. The minimum value of cosine**The minimum value of cosine is -1 , which occurs when

$$\frac{x}{2} = 180^\circ$$

$$x = 360$$

6. Read the graph

So

$$B = (360, -1)$$

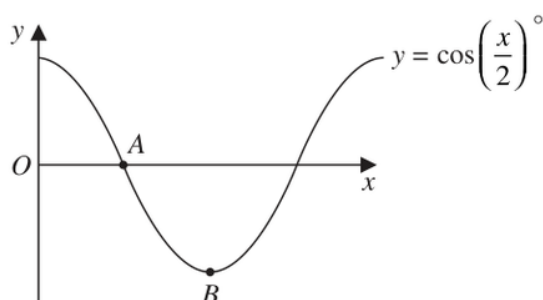
FINAL ANSWER

$$A = (180, 0), B = (360, -1)$$

PAST PAPER QUESTION**Q#: 23** **Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

23 The diagram shows a sketch of the graph of $y = \cos\left(\frac{x}{2}\right)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

(Total for Question 23 is 2 marks)

PAST PAPER SOLUTION**Q#: 23****Marks: 2****TOPIC: Graphs of Functions**

Jun 2022 P2H - Jun 2022 | P2H

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The graph is

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$$\cos\left(\frac{x}{2}\right) = 0$$

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3. Read the graph

So

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$$\frac{x}{2} = 180^\circ$$

$$x = 360$$

6. Read the graph

So

$$B = (360, -1)$$

FINAL ANSWER

$$A = (180, 0), B = (360, -1)$$

PAST PAPER QUESTION

Q#: 24

Marks: 3

TOPIC: **Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

24

$$\frac{18 \times (\sqrt{27})}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

(Total for Question 24 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

2. Use index laws

First,

$$\frac{18}{6} = 3 = 3^1$$

3. Simplify surd

Also,

$$\sqrt{27} = \sqrt{3^3} = 3^{3/2}$$

4. Simplify surd

So

$$(\sqrt{27})^{4n+6} = 3^{\frac{3}{2}(4n+6)} = 3^{6n+9}$$

5. Use index laws

And

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

6. Use index laws

Therefore the whole expression is

$$3^{1+6n+9-(4n+16)} = 3^{2n-6}$$

7. Use index laws

So

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6.$$

PAST PAPER QUESTION**Q#: 24** **Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

24

$$\frac{18 \times (\sqrt{27})}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

(Total for Question 24 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 24****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jun 2022 P2H - Jun 2022 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$\frac{18(\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

2. Use powers of 3

Use powers of 3:

$$18 \div 6 = 3$$

$$\sqrt{27} = (3^3)^{1/2} = 3^{3/2}$$

$$(\sqrt{27})^{4n+6} = 3^{6n+9}$$

$$9^{2n+8} = (3^2)^{2n+8} = 3^{4n+16}$$

3. Evaluate fraction

So

$$\frac{3 \cdot 3^{6n+9}}{3^{4n+16}} = 3^{1+6n+9-4n-16} = 3^{2n-6}$$

4. Calculate value

Therefore,

$$x = 2n - 6$$

FINAL ANSWER

$$x = 2n - 6$$